Electrons in disorder potential.

- Single particle problem in random $U(\vec{r}) = \sum_{i=1}^{N_{imp}} V(\vec{r}-\vec{R}_i)$.
- Single particle GF

$$G(r_1,\tau_1; r_2,\tau_2; U) = -\langle T_\tau \hat{\psi}(r_1\tau_1)\hat{\psi}^+(r_2\tau_2)\rangle$$

satisfies

$$\Big\{\frac{\partial}{\partial\tau_1} + \underbrace{\Big[-\frac{\vec{\nabla}_1^2}{2m} - \mu + \tilde{U}_0(\vec{r}_1)\Big]}_{\substack{\text{regular}\\ \text{(non-random)}\\ \text{part of the}\\ \text{single-particle}\\ \text{Hamiltonian, } \hat{h}_0(\vec{r}_1)}} + U(\vec{r}_1)\Big\} G(r_1\tau_1; r_2\tau_2; U) = \underbrace{-\delta(\tau_1-\tau_2)\,\delta(\vec{r}_1-\vec{r}_2)}_{\substack{\text{RHS (right-hand-side)}\\ \text{comes from } \frac{\partial}{\partial\tau_1} \text{ acting on } \hat{T}_\tau.}} \quad (1)$$

Short-hand notation: $G(r_1\tau_1; r_2\tau_2; U) = G(1,2;U)$

Think of random $U(r)$ as a perturbation. In its absence, the Hamiltonian is just $H_0 = \sum_i -\frac{\vec{\nabla}_i^2}{2m} - \mu + \tilde{U}_0(\vec{r}_i)$, and GF satisfies

$$\Big\{\frac{\partial}{\partial\tau_1} + \underbrace{\Big[-\frac{\vec{\nabla}_1^2}{2m} - \mu + \tilde{U}_0(\vec{r}_1)\Big]}_{\hat{h}_0(\vec{r}_1)}\Big\} G_0(r_1\tau_1; r_2\tau_2) = -\delta(\tau_1-\tau_2)\,\delta(\vec{r}_1-\vec{r}_2) \quad (2)$$

From (1) and (2) follows that

$$G(1,2;U) = G_0(1,2) + \int_0^\beta d\tau_3 \int d\vec{r}_3\, G_0(1,3)\, U(\vec{r}_3)\, G(3,2;U) \quad (3)$$

Proof: act on both sides of (3) with operator

$$\Big[\frac{\partial}{\partial\tau_1} + \hat{h}_0(\vec{r}_1)\Big]$$

$$\left[\frac{\partial}{\partial \tau_1} + \hat{h}_0(r_1)\right] G(1,2;u) = -\delta(\tau_1-\tau_2)\,\delta(r_1-r_2) +$$

$$+ \int d\tau_3\, d\vec{r}_3 \underbrace{\left[\frac{\partial}{\partial \tau_1} + \hat{h}_0(r_1)\right] G_0(1,3)}_{-\delta(\tau_1-\tau_3)\,\delta(\vec{r}_1-\vec{r}_3)} u(\vec{r}_3)\, G(3,2;u) =$$

$$= -\delta(\tau_1-\tau_2)\,\delta(r_1-r_2) - u(r_1)\, G(r_1\tau_1; r_2\tau_2; u)$$

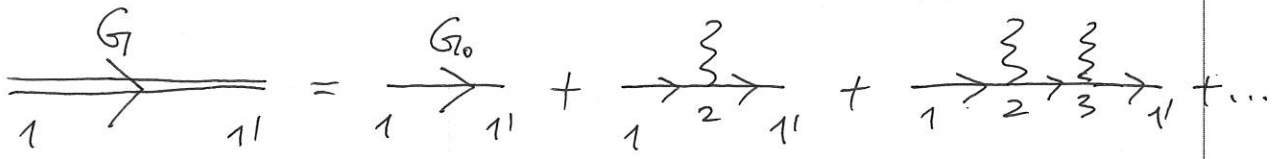
which is just eqn. (1).

Thus, we arrive at eqn. (4.23) of the textbook

$$G(1,1';u) = G_0(1,1') + \int d2\; G_0(1,2)\, u(2)\, G_0(2,1') +$$

$$+ \int d2\, d3\; G_0(1,2)\, u(2)\, G_0(2,3)\, u(3)\, G_0(3,1') + \dots \quad \left(\begin{array}{c}\text{expanded}\\ \text{version of}\\ (3)\end{array}\right) =$$

$$= G_0(1,1') + \int d2\; G_0(1,2)\, u(2)\, G(2,1;u) \qquad \left(\begin{array}{c}\text{compact version}\\ \text{of (3)}\end{array}\right).$$



For a given realization of disorder $u(\vec{r}) = \sum_i V(\vec{r}-\vec{R}_i)$
(i.e. given set of $\{\vec{R}_i\}$)

$G(1,1')$ is <u>not</u> translationally invariant.

Its <u>ensemble</u> <u>average</u> $\langle G(1,1')\rangle$ should, however, be translationally invariant. This is because averaging over u (i.e., over positions $\vec{R}_i$ of impurities) removes $\{\vec{R}_i\}$ dependence, and $G_0(1,1')$ <u>is</u> invariant.

Eqn for averaged $\langle G \rangle$

$$\langle G(1,1';U)\rangle = G_0(1,1') + \int d2\, G_0(1,2)\langle U(2)\rangle G_0(2,1')$$

$$+ \int d2\,d3\; G_0(1,2)\, G_0(2,3)\, G_0(3,1')\, \langle U(2)\, U(3)\rangle + \dots \quad (4)$$

Here $\langle U(2)\rangle = \langle \sum_i v(\vec{r}_2 - \vec{R}_i)\rangle = \langle \sum_i \sum_{\vec{k}} v(\vec{k})\, e^{i\vec{k}\cdot(\vec{r}_2-\vec{R}_i)}\rangle$

$v(\vec{k})$ is F.T. of $V(\vec{r})$

$$= \sum_i \sum_{\vec{k}} v(\vec{k})\, e^{i\vec{k}\cdot\vec{r}_2} \underbrace{\frac{1}{V}\int d^3\vec{R}_i\; e^{-i\vec{k}\cdot\vec{R}_i}}_{\langle e^{-i\vec{k}\cdot\vec{R}_i}\rangle} =$$

$$= \sum_i \sum_{\vec{k}} v(\vec{k})\, e^{i\vec{k}\cdot\vec{r}_2}\, \delta_{\vec{k},0} = \sum_i v(\vec{k}=0) = N_{imp}\, v(0)$$

This is just the average of random potential.

$$\langle U(2)\, U(3)\rangle = \langle \sum_{i,j} V(\vec{r}_2 - \vec{R}_i)\, V(\vec{r}_3 - \vec{R}_j)\rangle =$$

$$= \sum_{k_1,k_2} v(k_1)\, v(k_2) \langle \sum_{i,j} e^{i\vec{k}_1\cdot(\vec{r}_2-\vec{R}_i)}\, e^{i\vec{k}_2\cdot(\vec{r}_3-\vec{R}_j)}\rangle$$

$$\sum_{i,j} \langle e^{-i\vec{k}_1\cdot\vec{R}_i}\, e^{-i\vec{k}_2\cdot\vec{R}_j}\rangle = \underbrace{\sum_{i\neq j} \langle\ \rangle}_{\text{scattering off different impurities}} + \underbrace{\sum_{i=j} \langle\ \rangle}_{\text{double scattering from the same impurity}}$$

$$\langle U(2)\rangle = \sum_i \langle v(\vec{r}_2 - \vec{R}_i)\rangle$$

denotes average with probability distribution

$$P(\{\vec{R}_i\}) = V^{-N_{imp}}$$

i.e. every one of N_{imp} impurities is distributed uniformly over the volume V of the sample. In other words, probability to find impurity in small volume ΔV is $p = \frac{\Delta V}{V}$.

Then $\langle v(\vec{r} - \vec{R}_j)\rangle = \int \prod_{i=1}^{N_{imp}} d\vec{R}_i \; v(\vec{r} - \vec{R}_j)\, P(\{\vec{R}_i\})$

$$= \underbrace{\left(\prod_{i\neq j} \int \frac{d\vec{R}_i}{V}\right)}_{=1} \cdot \int \frac{d\vec{R}_j}{V}\, v(\vec{r} - \vec{R}_j)$$

$$\Rightarrow \langle U(2)\rangle = \sum_i \frac{1}{V} \int d\vec{R}_j \; v(\vec{r}_2 - \vec{R}_j) = \frac{N_{imp}}{V} \int d\vec{R}_j \; v(\vec{r}_2 - \vec{R}_j)$$

$$\cancel{= \rho_i \int d\vec{R}_j \sum_{\vec{k}} v(\vec{k})\, e^{i\vec{k}\cdot(\vec{r} - \vec{R}_j)} =}$$

$$\cancel{= \rho_i (2\pi)^3}$$

Suppose <u>short-ranged</u> impurity potential $v(\vec{r}) = v_0\, \delta(\vec{r})$

$$\Rightarrow \langle U(2)\rangle = \rho_i \int d\vec{R}\; v_0\, \delta(\vec{r}_2 - \vec{R}) = \rho_i\, v_0$$

where $\rho_i = \frac{N_{imp}}{V} \ll 1$ is concentration of impurities in the material.

$$\langle U(\vec{r}_1)\, U(\vec{r}_2)\rangle = \sum_{i,j} \langle v(\vec{r}_1-\vec{R}_i)\, v(\vec{r}_2-\vec{R}_j)\rangle$$

$$= \sum_{k_1,k_2} v(\vec{k}_1)\, v(\vec{k}_2) \sum_{i,j} e^{i\vec{k}_1\cdot\vec{r}_1 + i\vec{k}_2\cdot\vec{r}_2} \left\langle e^{-i\vec{k}_1\vec{R}_i - i\vec{k}_2\vec{R}_j}\right\rangle$$

$$= \sum_{k_1,k_2} v(\vec{k}_1)\, v(\vec{k}_2) \Big\{ \sum_{i=j} e^{i\vec{k}_1\vec{r}_1 + i\vec{k}_2\vec{r}_2} \langle e^{-i(\vec{k}_1+\vec{k}_2)\vec{R}_i}\rangle$$

$$+ \sum_{i\neq j} e^{i\vec{k}_1\vec{r}_1 + i\vec{k}_2\vec{r}_2} \langle e^{-i\vec{k}_1\vec{R}_i}\rangle_{R_i} \langle e^{-i\vec{k}_2\vec{R}_j}\rangle_{R_j} \Big\}$$

$$= \sum_{k_1,k_2} v(\vec{k}_1)\, v(\vec{k}_2) \Big\{ e^{i\vec{k}_1(\vec{r}_1-\vec{r}_2)}\, \rho_i\, (2\pi)^3 \delta(\vec{k}_1+\vec{k}_2)$$

$$+ \rho_i^2 (2\pi)^3 \delta(\vec{k}_1)\, (2\pi)^3 \delta(\vec{k}_2) \Big\}$$

Since $\langle U(\vec{r})\rangle = \sum_k v(\vec{k})\, \rho_i (2\pi)^3 \delta(\vec{k})$, we see that

$$\langle U(\vec{r}_1)\, U(\vec{r}_2)\rangle = \sum_{k_1 k_2} v(k_1)\, v(-k_1)\, \rho_i (2\pi)^3 \delta(k_1+k_2)\, e^{i\vec{k}_1(\vec{r}_1-\vec{r}_2)}$$

$$+ \langle U(\vec{r}_1)\rangle \langle U(\vec{r}_2)\rangle .$$

For short-ranged impurity $v(\vec{r}) = v_0\, \delta(\vec{r}) = v_0 \int \frac{d\vec{k}}{(2\pi)^3} e^{i\vec{k}\cdot\vec{r}} =$

$$= \int \frac{d\vec{k}}{(2\pi)^3} v(\vec{k})\, e^{i\vec{k}\cdot\vec{r}} \Rightarrow v(\vec{k}) = v_0$$

(all harmonics are equal)

Then $\sum_{k_1 k_2} |v(k_1)|^2 \rho_i (2\pi)^3 \delta(\vec{k}_1+\vec{k}_2)\, e^{i\vec{k}_1(\vec{r}_1-\vec{r}_2)}$

$$= \int \frac{d\vec{k}_2}{(2\pi)^3} \int \frac{d\vec{k}_1}{(2\pi)^3} \rho_i v_0^2 (2\pi)^3 e^{i\vec{k}_1(\vec{r}_1-\vec{r}_2)}\, \delta(\vec{k}_1+\vec{k}_2) =$$

$$= \rho_i v_0^2\, \delta(\vec{r}_1-\vec{r}_2)$$

Hence $\langle U(\vec{r}_1)U(\vec{r}_2)\rangle = \rho_i v_0^2\,\delta(\vec{r}_1-\vec{r}_2) + (\rho_i v_0)^2$

$$\langle U(\vec{r}_1)\rangle = \rho_i v_0$$

for a delta-function potential $v(\vec{r}) = v_0\,\delta(\vec{r})$.

Assuming small impurity concentration, $\rho_i = \frac{N_{imp}}{V} \ll 1$, and weak potential $v_0 \to 0$, we can restrict ourselves by terms linear in ρ_i.

$$\sum_{i\neq j} \langle \quad \rangle = \sum_i \langle e^{-i\vec{k}_1\cdot\vec{R}_i} \rangle \sum_j \langle e^{-i\vec{k}_2\cdot\vec{R}_j} \rangle =$$

$$= \sum_i \frac{1}{V}\int d^3\vec{R}_i\, e^{-i\vec{k}_1\cdot\vec{R}_i} \cdot \sum_j \frac{1}{V}\int d^3\vec{R}_j\, e^{-i\vec{k}_2\cdot\vec{R}_j} =$$

$$= \sum_i \delta_{\vec{k}_1,0} \sum_j \delta_{\vec{k}_2,0}$$

$$\sum_{i=j} \langle \quad \rangle = \sum_i \langle e^{-i(\vec{k}_1+\vec{k}_2)\cdot\vec{R}_i} \rangle = \sum_i \frac{1}{V}\int d^3\vec{R}_i\, e^{-i(\vec{k}_1+\vec{k}_2)\cdot\vec{R}_i}$$

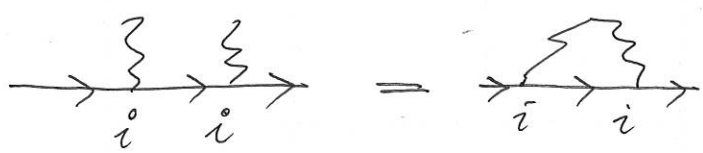
$$= \sum_i \delta_{\vec{k}_1+\vec{k}_2,0}$$

$$\Rightarrow \langle u(2)\, u(3) \rangle = \sum_{k_1,k_2} v(k_1)\, v(k_2) \sum_{i,j} \delta_{\vec{k}_1,0}\, \delta_{\vec{k}_2,0}$$

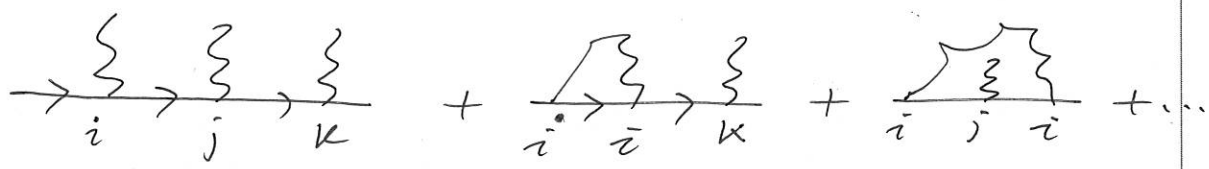
$$+ \sum_{k_1,k_2} v(k_1)\, v(k_2) \sum_i \delta_{k_1+k_2,0} = N_{imp}^2 \left(v(0)\right)^2$$

$$+ N_{imp} \sum_{\vec{k}_1} v(\vec{k}_1)\, v(-\vec{k}_1)$$

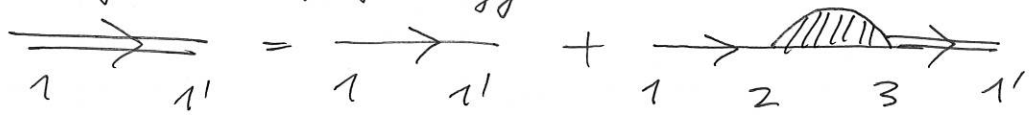
$\Rightarrow$



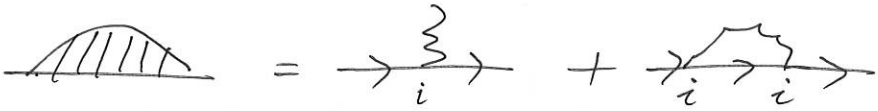
We now select in the infinite series (4) all terms describing scattering off single impurity ($\langle U(2)\rangle$) <u>and</u> double scattering of the same impurity:



$\Rightarrow$ series for the self-energy:



$$\langle G(1,1')\rangle = G_0(1,1') + \int d2\,d3\; G_0(1,2)\,\Sigma(2,3)\,\langle G(3,1')\rangle$$

$\Sigma(2,3) =$ 

The averaged $\langle G\rangle$ is translationally invariant

$$\Rightarrow \quad G(\vec{p},\omega_n) = G_0(\vec{p},\omega_n) + G_0(\vec{p},\omega_n)\,\Sigma(\vec{p},\omega_n)\,G(\vec{p},\omega_n)$$

$$G = G_0 + G_0\,\Sigma\,G$$

$$G\,(1 - G_0\,\Sigma) = G_0$$

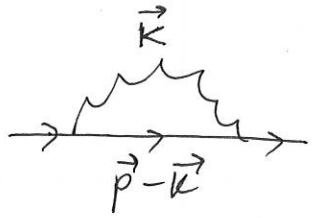
$$G = \frac{G_0}{1 - G_0\,\Sigma} = \frac{1}{G_0^{-1} - \Sigma}$$

$$G(\vec{p},\omega_n) = \frac{1}{G_0^{-1}(\vec{p},\omega_n) - \Sigma(\vec{p},\omega_n)}$$

$$G_0(\vec{p}, \omega_n) = \frac{1}{i\omega_n - \xi(\vec{p})}$$

$$\Sigma = \underset{\substack{\uparrow \\ \text{single} \\ \text{scattering}}}{\Sigma^{(1)}} + \underset{\nwarrow \text{ double scattering}}{\Sigma^{(2)}}$$

$$\Sigma^{(1)}(\vec{p}, \omega_n) = N_{imp}\, v(0)$$ is a constant shift which can be absorbed into chemical potential.

$\Sigma^{(2)}(\vec{p}, \omega_n) =$ 

$=$

$$= N_{imp} \sum_{\vec{k}} |v(\vec{k})|^2 G_0(\vec{p}-\vec{k}, \omega_n) =$$

$$= \left(\vec{k} = \vec{p} - \vec{p}\,', \Rightarrow \vec{p}\,' = \vec{p} - \vec{k}\right)$$

$$= N_{imp} \sum_{\vec{p}\,'} |v(\bar{p}-\bar{p}')|^2 G_0(\vec{p}\,', \omega_n) =$$

$$= N_{imp} \sum_{\vec{p}\,'} |v(\bar{p}-\bar{p}')|^2 \frac{1}{i\omega_n - \xi(\bar{p}')}$$

$$= -N_{imp} \sum_{\vec{p}\,'} |v(\bar{p}-\bar{p}')|^2 \frac{i\omega_n + \xi(\vec{p}\,')}{\omega_n^2 + \xi^2(\vec{p}\,')}$$

For delta-functional impurities (very short-ranged)

$$V(\vec{r}) = v_0\, \delta(\vec{r})$$

$$\Rightarrow v(\vec{q}) = \int V(\vec{r})\, e^{-i\vec{q}\cdot\vec{r}}\, d\vec{r} = v_0$$

$$\Sigma^{(2)}(\vec{p}, \omega_n) = -N_{imp} \int \frac{d\vec{p}\,'}{(2\pi)^3} \left|U(\vec{p}-\vec{p}\,')\right|^2 \frac{i\omega_n + \xi(\vec{p}\,')}{\omega_n^2 + \xi^2(\vec{p}\,')}$$

$$\cong -N_{imp}\, v_0^2\, \frac{4\pi p_F^2}{(2\pi)^3 v_F} \int_{-\infty}^{\infty} d\xi' \frac{i\omega_n + \xi'}{\omega_n^2 + (\xi')^2}$$

$$\int d\vec{p}\,' = \int_0^{\infty} dp\, p^2 \underbrace{\int_0^{2\pi} d\varphi \int_0^{\pi} d\theta}_{4\pi} \approx \frac{4\pi p_F^2}{v_F} \underbrace{\int d(v_F p)}_{\int d\xi}$$

$$\int_{-\infty}^{\infty} d\xi \frac{i\omega_n + \xi}{\omega_n^2 + \xi^2} = \frac{i\omega_n}{|\omega_n|} \tan^{-1}\left(\frac{\xi}{|\omega_n|}\right)\Bigg|_{-\infty}^{\infty} = \pi \frac{i\omega_n}{|\omega_n|}$$

$$\Sigma^{(2)}(\vec{p}, \omega_n) = -\frac{N_{imp}\, v_0^2}{(2\pi)} \frac{p_F^2}{v_F} \frac{i\omega_n}{|\omega_n|} \equiv -\frac{i\omega_n}{|\omega_n|} \frac{1}{2\tau}$$

$$\Rightarrow G(\vec{p}, \omega_n) = \frac{1}{i\omega_n - \xi(\vec{p}) + \frac{i\omega_n}{|\omega_n| 2\tau}}$$

$\Rightarrow$ for $\omega_n > 0$ $\quad \frac{i\omega_n}{|\omega_n|} = i$

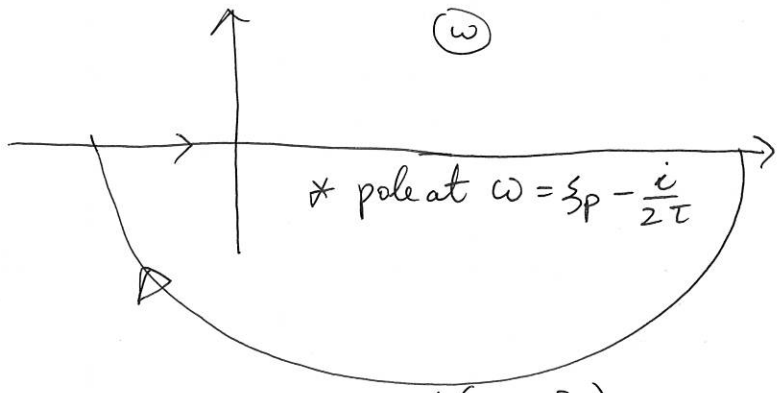
$$G(\vec{p}, \omega_n) = \frac{1}{i\omega_n - \xi_p + \frac{i}{2\tau}} \quad \text{for } \omega_n > 0.$$

Hence for the <u>retarded</u> GF

$$G_R(\vec{p},\omega) = G(\vec{p}, i\omega_n = \omega) =$$

$$= \frac{1}{\omega - \xi_p + \frac{i}{2\tau}}$$

$$G_R(\vec{p},t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{e^{-i\omega t}}{\omega - \xi_p + \frac{i}{2\tau}}$$



$e^{-i\omega t} =$
$= e^{-i(\omega' + i\omega'')t}$
$= e^{-i\omega' t + \omega'' t}$

$\Rightarrow$ for $t > 0$ need to close integration contour in $\omega'' < 0$ half-plane

$$\Rightarrow G_R(\vec{p},t) = -\frac{2\pi i}{2\pi} e^{-it(\xi_p - \frac{i}{2\tau})}$$

$$= -i\,\theta(t)\, e^{-i\xi(p)t}\, e^{-\frac{t}{2\tau}}$$

decays exponentially in time

Also, $$G_R(\vec{r},\omega) = \int \frac{d^3\vec{p}}{(2\pi)^3} \frac{e^{i\vec{p}\cdot\vec{r}}}{\omega - \xi(p) + \frac{i}{2\tau}} =$$

$$= -\frac{m}{2\pi r} e^{ir(p_F + \frac{\omega}{v_F})} e^{-\frac{r}{2l}}, \quad l = v_F \tau$$

decays exponentially in space.

$$G_R(\vec{r},\omega) = \int \frac{dp\, p^2\, d\varphi\, \sin\theta\, d\theta}{(2\pi)^3} \frac{e^{ipr\cos\theta}}{\omega - \xi(p) + \frac{i}{2\tau}} =$$

(do θ integral)

$$= \frac{2\pi}{(2\pi)^3} \int_0^\infty dp\, p^2 \int_{-1}^{1} d(\cos\theta) \frac{e^{ipr\cos\theta}}{\omega - \xi(p) + \frac{i}{2\tau}} =$$

$$= \frac{1}{(2\pi)^2} \int_0^\infty dp\, p^2 \frac{2i \sin pr}{ipr\left(\omega - \xi(p) + \frac{i}{2\tau}\right)} =$$

$$= \frac{-i}{4\pi^2 r} \int_0^\infty dp\, p \frac{e^{ipr} - e^{-ipr}}{\omega - \xi(p) + \frac{i}{2\tau}}$$

$$\xi(p) = \frac{\vec{p}^{\,2}}{2m} - \frac{\vec{p}_F^{\,2}}{2m} = \frac{(\vec{p}+\vec{p}_F)(\vec{p}-\vec{p}_F)}{2m} = v_F(p - p_F)$$ (energy depends on "radial" component of $\vec{p}$)

$$\Rightarrow \quad p = p_F + \frac{\xi}{v_F}$$

$$G(\vec{r},\omega) \simeq \frac{-i p_F}{4\pi^2 r v_F} \int_{-\infty}^{\infty} d\xi \frac{e^{i(p_F + \frac{\xi}{v_F})r} - e^{-i(p_F + \frac{\xi}{v_F})r}}{\omega - \xi + \frac{i}{2\tau}}$$

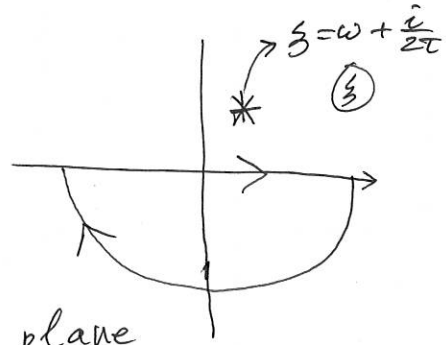
pole at $\xi = \omega + \frac{i}{2\tau}$

here $r = |\vec{r}| > 0$

$e^{i\xi r} = e^{i(\xi' + i\xi'')r} = e^{i\xi' r} e^{-\xi'' r} \Rightarrow$ $e^{i\xi r}$ integral converges for $\xi'' > 0$

$$I_1 = \int_{-\infty}^{\infty} d\xi \frac{e^{i\frac{\xi r}{v_F}}}{\xi - \omega - \frac{i}{2\tau}} = 2\pi i\, e^{i\frac{r}{v_F}(\omega + \frac{i}{2\tau})} = 2\pi i\, e^{i\frac{r\omega}{v_F}} e^{-\frac{r}{2v_F\tau}}$$

$$I_2 = \int_{-\infty}^{\infty} d\xi \frac{e^{-i\frac{\xi r}{v_F}}}{\xi - \omega - \frac{i}{2\tau}} \longrightarrow$$



need to close contour in $\xi'' < 0$ plane which contains no pole $\Rightarrow I_2 = 0$

Thus $G_R(\vec{r},\omega) = \dfrac{i p_F}{4\pi^2 r v_F} e^{i p_F r} \displaystyle\int_{-\infty}^{\infty} d\xi \, \frac{e^{i\xi \frac{r}{v_F}}}{\xi - \omega - \frac{i}{2\tau}}$

$$= \frac{i p_F}{4\pi^2 r v_F} (2\pi i) \, e^{i p_F r} e^{i \frac{r\omega}{v_F}} e^{-\frac{r}{2 v_F \tau}} =$$

$$= - \frac{p_F}{2\pi v_F} \frac{e^{i\left(p_F + \frac{\omega}{v_F}\right) r} e^{-\frac{r}{2 v_F \tau}}}{r}$$

decays exponentially $\sim e^{-\frac{r}{2l}}$, where $l = v_F \tau$ is the mean free path for electron at the Fermi surface

<u>Keldysh model</u> represents an opposite limit: instead of very short-ranged disorder we consider a <u>long-ranged correlated</u> one

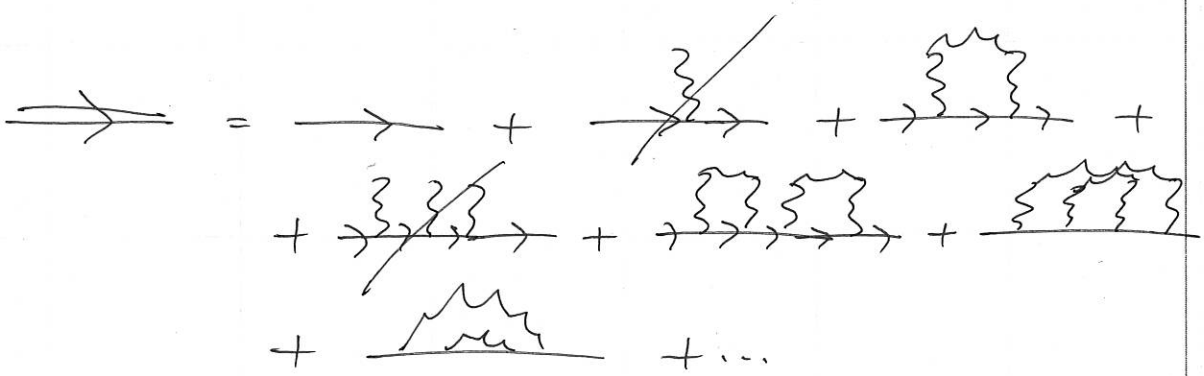
$$V(\vec{r}) = \sum_i v(\vec{r}-\vec{R}_i)$$

$\langle V(\vec{r})\rangle = 0$ (zero on average)

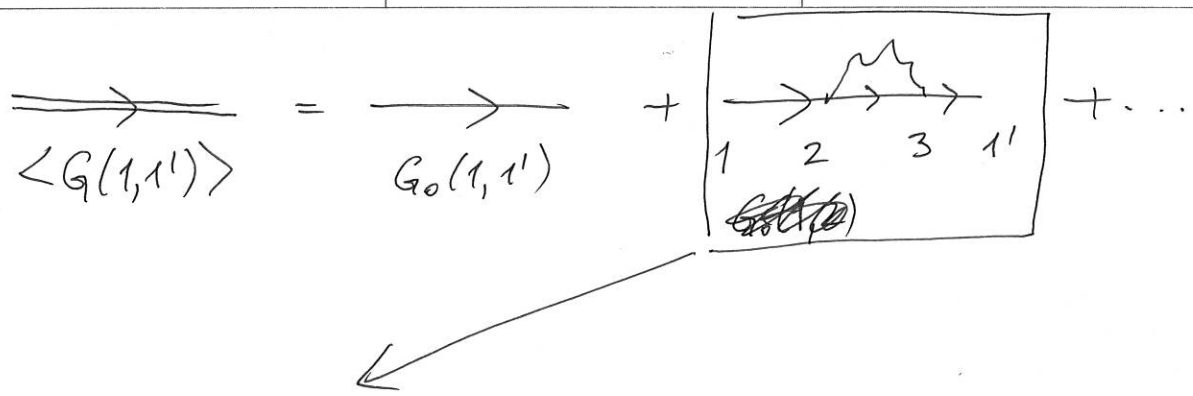
$\langle V(\vec{r}_1)\, V(\vec{r}_2)\rangle = W^2$ (constant $W^2 \Rightarrow$ disordered potential at $\vec{r}_1$ is correlated with that at $\vec{r}_2$, no matter how far $\vec{r}_{1,2}$ are from each other)

$$\langle V(\vec{r}_1)\, V(\vec{r}_2)\rangle = W^2 = \int \frac{d\vec{q}}{(2\pi)^3}\, e^{i\vec{q}(\vec{r}_1-\vec{r}_2)} f(\vec{q})$$

Since W^2 does not depend on $\vec{r}_1-\vec{r}_2$, $\Rightarrow f(\vec{q}) = (2\pi)^3 W^2 \delta(\vec{q})$



After averaging over the disorder, the series only include terms with <u>even</u> number $V(\vec{r})$ which are <u>paired</u> to each other in all possible ways.



$$\int d2\,d3\; G_0(1,2)\, G_0(2,3)\, G_0(3,1')\, \underbrace{\langle V(2)\, V(3)\rangle}_{W^2} =$$

$$= W^2 \int d\tau_2\, d\vec{r}_2 \int d\tau_3\, d\vec{r}_3\; G_0(\vec{r}_2-\vec{r}_1, \tau_2-\tau_1)\, G_0(\vec{r}_3-\vec{r}_2, \tau_3-\tau_2)\, G_0(\vec{r}_3-\vec{r}_1', \tau_3-\tau_1')$$

F.T. $G_0(\vec{r}, t) = \int \frac{d\vec{k}}{(2\pi)^3}\, e^{i\vec{k}\cdot\vec{r} - i\omega_n \tau}\, G_0(\vec{k}, \omega_n)$

$\Rightarrow$ we get $\int_{(\vec{k}_1, \vec{k}_2, \vec{k}_3)} W^2 \int d\tau_2\, d\vec{r}_2 \int d\tau_3\, d\vec{r}_3\; e^{i\vec{k}_1(\vec{r}_2-\vec{r}_1) + i\vec{k}_2(\vec{r}_3-\vec{r}_2) + i\vec{k}_3(\vec{r}_1'-\vec{r}_3)}$

$$\cdot\, e^{-i\omega_1(\tau_2-\tau_1) - i\omega_2(\tau_3-\tau_2) - i\omega_3(\tau_1'-\tau_3)}$$

$$\cdot\, G_0(\vec{k}_1, \omega_1)\, G_0(\vec{k}_2, \omega_2)\, G_0(\vec{k}_3, \omega_3)$$

but $\int d\vec{r}_2\, e^{i\vec{r}_2(\vec{k}_1-\vec{k}_2)} \int d\vec{r}_3\, e^{i\vec{r}_3(\vec{k}_2-\vec{k}_3)} =$

$$= (2\pi)^3\, \delta(\vec{k}_1-\vec{k}_2)\, (2\pi)^3\, \delta(\vec{k}_2-\vec{k}_3)$$

and the same for τ-integrals.

Hence the whole term gives

$$W^2 \int \frac{d\vec{k}_1}{(2\pi)^3}\, \frac{1}{\beta} \sum_{\omega_1} e^{i\vec{k}_1(\vec{r}_1'-\vec{r}_1) - i\omega_1(\tau_1'-\tau_1)} \cdot$$

$$\cdot\, G_0(\vec{k}_1, \omega_1)\, G_0(\vec{k}_1, \omega_1)\, G_0(\vec{k}_1, \omega_1)$$

Clearly this works for every term in the series for $\langle G(1,1') \rangle \Rightarrow$ can write it in $(\vec{k},\omega)$-space

$$G(\vec{k},\omega) = G_0(\vec{k},\omega) + A_1 W^2 G_0^3(\vec{k},\omega) + A_2 (W^2)^2 G_0^5(\vec{k},\omega) + \ldots$$

$$= G_0(\vec{k},\omega)\left[1 + \sum_{n=1}^{\infty} A_n (W^2)^n G_0^{2n}(\vec{k},\omega)\right]$$

Here A_n = number of diagrams in a given, W^{2n}, order.

* Note that unlike the case of short-ranged disorder, here we sum <u>all</u> diagrams (not just self-energy ones)

<u>Calculate A_n</u>:

W^{2n} order: $2n$ wavy lines $\{$
$2n+1$ electron GF

1) Take 1st wavy line $\Rightarrow$ connect to any of $(2n-1)$ wavy lines

2) Out of remaining $(2n-2)$ lines pick one $\Rightarrow$ connect to $(2n-3)$

3) ... continue like this.

$\Rightarrow$ total number of ways to connect all wavy lines is $A_n = (2n-1)(2n-3)(2n-5)\ldots = (2n-1)!!$

Next observe that $\Gamma(n+\frac{1}{2}) = 1\cdot 3\cdot 5\cdot \ldots (2n-3)\cdot(2n-1)\frac{\sqrt{\pi}}{2^n}$

$$\Rightarrow (2n-1)!! = \frac{2^n}{\sqrt{\pi}}\,\Gamma(n+\tfrac{1}{2})$$

Next, $\Gamma(z) = \int_0^\infty dx\; x^{z-1} e^{-x}$

$$\Rightarrow \Gamma(n+\tfrac{1}{2}) = \int_0^\infty dx\; x^{n-\frac{1}{2}} e^{-x} = \int_0^\infty dx\; x^{\frac{2n-1}{2}} e^{-x}$$

let $x = \frac{1}{2}t^2$, $dx = t\,dt$

$$\Gamma(n+\tfrac{1}{2}) = \int_0^\infty dt\; t\left(\frac{t^2}{2}\right)^{\frac{2n-1}{2}} e^{-\frac{1}{2}t^2} =$$

$$= \int dt\,(t)\,t^{2n-1}\; e^{-\frac{1}{2}t^2}\; 2^{(\frac{1}{2}-n)}$$

$$\Rightarrow (2n-1)!! = \frac{\sqrt{2}}{\sqrt{\pi}}\int_0^\infty dt\; t^{2n}\, e^{-\frac{1}{2}t^2} = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dt\, t^{2n} e^{-\frac{1}{2}t^2}$$

Then $G(\vec{k},\omega) = G_0(\vec{k},\omega)\left\{1 + \sum_{n=1}^{\infty}\frac{1}{\sqrt{2\pi}}\int_0^\infty dt\; e^{-\frac{1}{2}t^2}\, t^{2n}\omega^{2n}\cdot G_0^{2n}(\vec{k},\omega)\right\}$

Note $\int_{-\infty}^{\infty} dt\; e^{-\frac{1}{2}t^2} = 2\sqrt{\frac{\pi}{2}} = \sqrt{2\pi}$.

Do the sum $\sum_{n=1}^{\infty} t^{2n}\omega^{2n}G_0^{2n} = t^2\omega^2 G_0^2 + t^4\omega^4 G_0^4 + \ldots =$

$$= (t^2\omega^2 G_0^2)(1 + t^2\omega^2 G_0^2 + t^4\omega^4 G_0^4 + \ldots) =$$

$$= \frac{t^2\omega^2 G_0^2}{1 - t^2\omega^2 G_0^2}$$

$$G(\vec{k},\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt \; e^{-\frac{1}{2}t^2} \; G_0(k,\omega) +$$

$$+ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt \; e^{-\frac{1}{2}t^2} \; G_0(k,\omega) \sum_{n=1}^{\infty} t^{2n} W^{2n} G_0^{2n}(k,\omega)$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt \; e^{-\frac{1}{2}t^2} \; G_0(k,\omega) \left\{ 1 + \frac{t^2 W^2 G_0^2(k,\omega)}{1 - t^2 W^2 G_0^2(k,\omega)} \right\}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt \; e^{-\frac{1}{2}t^2} \; \frac{G_0(k,\omega)}{1 - t^2 W^2 G_0^2(k,\omega)} =$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt \; e^{-\frac{1}{2}t^2} \; \frac{1}{2} G_0 \left(\frac{1}{1 - tWG_0} + \frac{1}{1 + tWG_0} \right)$$

($t \to -t$ in 2nd term)

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt \; e^{-\frac{1}{2}t^2} \; \frac{G_0(k,\omega)}{1 - tW G_0(k,\omega)} =$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt \; e^{-\frac{1}{2}t^2} \; \frac{1}{G_0^{-1}(\vec{k},\omega) - tW}$$

let now $t = \frac{V}{W}$

$$G(\vec{k},\omega_n) = \frac{1}{\sqrt{2\pi}\, W} \int_{-\infty}^{\infty} dV \; e^{-\frac{V^2}{2W^2}} \; \frac{1}{i\omega_n - \xi_p - V}$$

electron moves in a uniform potential V, value of which is distributed by Gauss law with width W.

Thus $G_R(\vec{k}, \varepsilon) = G(\vec{k}, i\omega_n = \varepsilon) =$

$$= \frac{1}{\sqrt{2\pi}\, W} \int_{-\infty}^{\infty} dV\, e^{-\frac{V^2}{2W^2}} \frac{1}{\varepsilon - \xi_k - V + i\delta}$$

Density of states

$$N(\varepsilon) = -\frac{1}{\pi} \int \mathrm{Im}\, G_R(\vec{k}, \varepsilon) \frac{d\vec{k}}{(2\pi)^3} = \frac{1}{\sqrt{2\pi}\, W} \int_{-\infty}^{\infty} dV\, e^{-\frac{V^2}{2W^2}} \cdot$$

$$\cdot \underbrace{\int \frac{d\vec{k}}{(2\pi)^3}\, \delta(\varepsilon - \xi_k - V)}$$

$$\frac{1}{2\pi^2} \int_0^{\infty} dk\, k^2\, \delta(\varepsilon - \xi_k - V) \approx \frac{k_F^2}{2\pi^2 v_F} \int_{-\infty}^{\infty} d\xi\, \delta(\varepsilon - V - \xi)$$

$$\xi = v_F k$$

$$\Rightarrow N(\varepsilon) \sim \frac{k_F^2}{v_F W} \int d\xi \int dV\, e^{-\frac{V^2}{2W^2}}\, \delta(\varepsilon - \xi - V)$$

$$= \frac{k_F^2}{v_F W} \int d\xi\, e^{-\frac{(\varepsilon - \xi)^2}{2W^2}}$$

$$\frac{1}{4\pi^2}\int_0^\infty dk^2 \, \delta\left(\varepsilon - V - \frac{k^2}{2m}\right) = \frac{(2m)^{3/2}}{4\pi^2}\sqrt{\varepsilon - V}\;\theta(\varepsilon - V)$$

$$k^2 = 2m(\varepsilon - V)$$

$$N(\varepsilon) = \frac{1}{\sqrt{2\pi}\,W}(2m)^{3/2}\int_{-\infty}^{\infty} dV\,\sqrt{\varepsilon - V}\;e^{-\frac{V^2}{2W^2}}\,\theta(\varepsilon - V)$$

$$= \frac{2\pi^{-1/2}m^{3/2}}{W}\int_{-\infty}^{\varepsilon} dV\,\sqrt{\varepsilon - V}\;e^{-\frac{V^2}{2W^2}}$$

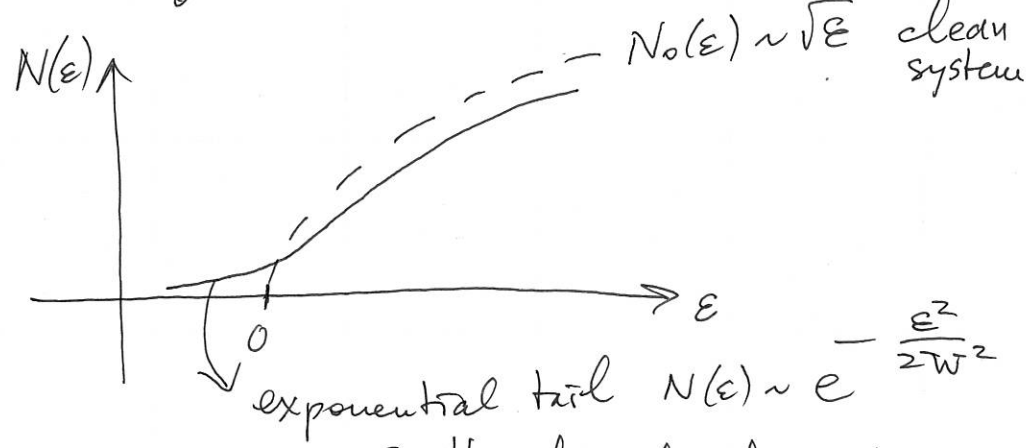
Important property: $N(\varepsilon) \neq 0$ for $\varepsilon < 0$ and $|\varepsilon| \gg W$

$\Rightarrow$ density of states acquires <u>exponential "tail"</u>

To see this, let $\varepsilon = -|\varepsilon|$ and $V = -x$:

$$N(\varepsilon) \sim \int_{|\varepsilon|}^{\infty} dx\,\sqrt{x - |\varepsilon|}\;e^{-\frac{x^2}{2W^2}}$$

Main contribution comes from vicinity of $x \sim |\varepsilon|$

$\Rightarrow x = |\varepsilon| + y$

$$N(\varepsilon) \sim \int_0^\infty dy\,\sqrt{y}\;e^{-\frac{(|\varepsilon| + y)^2}{2W^2}} \sim e^{-\frac{\varepsilon^2}{2W^2}}$$